Matrix Chain Multiplication

A_1 (10×100), A_2 (100×5) A_3 (5×50)

$((A_1 A_2) A_3)$ = 10.100.5 = 5000 Scalar multiplication
to Compute 10×5 Matrix (A_1A_2) +
10.5.50 = 2500 Scalar multiplication
to multiply this matrix by A_3
If we multiply as $(A_1 (A_2 A_3))$
= 100.5.50 = 25000 Scalar multiplication
to Compute 100×50 matrix Product $A_2 A_3$, +
10.100.50 = 50000 Scalar multiplication to multiply A_1 by
this matrix. for a total of 75,000 Scalar multiplication,
∴ Computing the Product according to first Paranthesis
is 10 times faster

The Goal is only to determine an order for multiplying matrices that has lowest cost.
Time in investing to determine this optimal order is more than for by the time saved later on when actually performing the matrix multiplication.

Counting the Number of Paranthesis.

$$P(n) = \begin{cases} 1 & \text{if } n = 1 \\ \sum_{k=1}^{n-1} P(k) \cdot P(n-k) & \text{if } n \geq 2 \end{cases}$$

Solution to this recurrence is $\Omega(2^n)$ thus it is exponential in n.

Apply Dynamic Programming

· We use dynamic Programming to determine how to optimize optimally Paranthesize a matrix chain.

optimal solution

1. Characterize the structure of an optimal solution
2. Recursively define the value of an optimal solution
3. Compute the value of an optimal Solution
4. Construct an optimal Solution from Computed Information

Step 1 The Structure of an optimal Paranthesization

Find the optimal Sub Structure and then use it to Construct an optimal solution to the Problem from optimal Solutions to Sub Problems.

$A_{i} \ldots A_{j}$ $(i<j)$

split $(i \leq k < j)$ (A_k and A_{k+1}) find product

$(A_{i \ldots k})$ and $(A_{k+1 \ldots j})$

cost of mul + cost of mul

optimally Paranthesizing

$(A_i A_{i+1} \ldots A_k$ and $A_{k+1} A_{k+2} \ldots A_j)$

Step 2 A. Recursive Solution :-

Matrix Chain multiplication

$A_1 \cdot A_2 \cdot A_3 \quad A_4$

$5\times4 \quad 4\times6 \quad 6\times2 \quad 2\times7$

let $A \cdot B = C$

$5\times\boxed{4 \quad 4}\times3 = 5\times3 = 15$

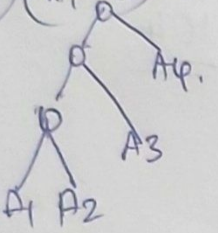
$5\times3\times4 = 60$ multiplications

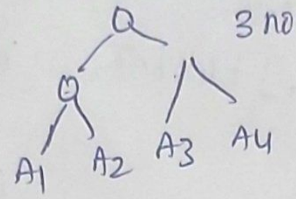
$A_1 \cdot A_2 \cdot A_3 \cdot A_4$

We can multiply any pair.
Question is how to multiply.

(1) $((A_1 A_2) A_3) A_4$

(2) $(A_1 A_2) \cdot (A_3 \cdot A_4)$





3 nodes

(3) $A_1 \cdot (A_2 \ A_3 \ A_4)$

(4) $A_1 \cdot (A_2 A_3) \cdot A_4$

~~$A_1 \cdot (A_2 \cdot (A_3 \cdot A_4))$~~

~~(3) $((A_1 A_2) \cdot A_3) \cdot A_4$~~

(5) $A_1 \cdot (A_2 (A_3 \cdot A_4))$

for N nodes $= \dfrac{2nC_n}{n+1}$ Trees are possible.

$T(3) = 5$

DP - Try all possibilities and pick up the best one

Dynamic Programming uses Tabulation method.
In Tabulation Method it uses bottom up approach

A_1 - not multiplied with any one

$\therefore M[1,1] = 0$

$A_2, \ M[2,2] = 0$

$A_3 \ \ M[3,3] = 0$

$A_4 \ \ M[4,4] = 0$

$m[1,2]$ - multiplying two matrices 1 to 2

$A_1 \quad A_2$

$5\times\boxed{4 \quad 4}\times6 \Rightarrow 5\times6\times4 = 120$

$m[2,3]$ - i.e. $A_2 \cdot A_3$

$4\times6 \quad 6\times2 \Rightarrow 4\times2\times6 = 48$

$m[3,4]$ - $A_3 \cdot A_4$

$6\times2 \quad 2\times7 = 6\times7\times2 = 84$

We have taken multiplication of pairs matrices

M	1	2	3	4
1	0	120	88	158
2		0	48	104
3			0	84
4				0

S	1	2	3	4
1		1	1	3
2			2	3
3				3
4				

$m[1,3] = \min \left[A_1 (A_2 \cdot A_3) \right. \qquad (A_1 A_2) \cdot A_3$

$= 5 \times 4 \quad 4 \times 6 \quad 6 \times 2 \qquad 5 \times 4 \quad 4 \times 6 \quad 6 \times 2$

results 1 to dimension of A_1

$= m[1,1] + m[2,3] + 5 \times 4 \times 2 \;, \quad m[1,2] + m[3,3] + 5 \times 6 \times 2$

$= (0 + 48 + 40 \;, \quad 120 + 0 + 60)$

$\min(88 \;, \; 180)$

$= 88$

$m[2,4] = A_2 (A_3 A_4) \qquad (A_2 A_3) A_4$

$4 \times 6 \quad 6 \times 2 \quad 2 \times 7 \qquad 4 \times 6 \quad 6 \times 2 \quad 2 \times 7$

$= m[2,2] + m[3,4] + 4 \times 6 \times 7 \qquad m[2,3] + m[4,4] + 4 \times 2 \times 7$

$= 0 + 84 + 168 \qquad 48 + 0 + 56$

$= \min(252 \;, \quad 104) = \underline{104}$

$A_1 \; 5 \times 4 \quad A_2 \; 4 \times 6 \quad A_3 \; 6 \times 2 \quad A_4 \; 2 \times 7$

$m[1,4] = \min \begin{bmatrix} m[1,1] + m[2,4] + 5 \times 4 \times 7 \;, \\ m[1,2] + m[3,4] + 5 \times 6 \times 7 \;, \\ m[1,3] + m[4,4] + 5 \times 2 \times 7 \end{bmatrix}$

$= \min \{ 0 + 104 + 140 \;, \quad 120 + 84 + 210 \;, \quad 88 + 0 + 70 \}$

$= \min \{ 244 \;, \quad 414 \;, \quad 158 \}$

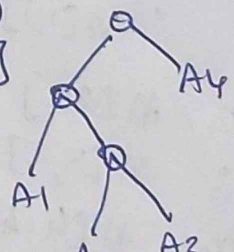
$= \underline{158}$

$A_1 \; 5 \times 4 \quad A_2 \; 4 \times 6 \quad A_3 \; 6 \times 2 \quad A_4 \; 2 \times 7$

$d_0 \; d_1 \quad d_2 \quad d_3 \quad d_4$

formula

$m[i,j] = \min \{ m[i,k] + m[k+1, j] + d_{i-1} * d_k * d_j \}$



$(A_1 \; A_2 \; A_3)(A_4)$

$((A_1)(A_2 \cdot A_3)) \cdot (A_4)$

Time Complexity $= \frac{n(n+1)}{2} = n^2 * n$ How many times we compute.

$O(n^3)$